

MaskMaker Usage Guide (Unity 2022.3)

This document explains how to use **MaskMaker** ([Tools > MaskMaker](#)). It is a tool that allows you to visually select UV islands of a mesh in the Unity Editor and export them as a mask image (PNG) or vertex colors.

Key Features

- **UV Island Selection:** Select directly by clicking the mesh in the Scene View or the Preview area.
 - **Hand-painting:** Use the brush tool on the preview to freely paint masks. Can be combined with island selection for merged output.
 - **Mask Image Import:** Load an existing mask image and use its black regions as hand-painted areas, allowing you to resume editing from a previously exported mask.
 - **Automatic UV Analysis:** UV layout is automatically analyzed when the target is set, allowing you to start working immediately.
 - **Work Copy Function:** Automatically creates a static working copy to prevent misalignment due to mesh deformation.
 - **Flexible Export:** Supports channel-specific writing (RGBA) and baking to vertex colors.
 - **Automated Import Settings:** Exported textures automatically have [Read/Write Enabled](#) and [Streaming Mipmaps](#) enabled.
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Basic Steps

STEP 1: Set Target Model

1. Open [Tools > MaskMaker](#) from the menu.
2. Drag and drop the target GameObject into the [Target Model](#) frame.
3. Select the **Target Material** (if extracting from a specific submesh) and **UV Channel** (usually UV0).
 - *Note: UV analysis is executed automatically when the target is set.*
 - *Note: A working copy ([\[WorkCopy\]](#)) is created by default.*
4. (Optional) To start from an existing mask, drag and drop the mask image PNG into the [Import Mask Image](#) card and click **Load Mask**. See [Import Mask Image](#) for details.

STEP 2: Island Selection & Painting

1. Click the mesh in the Scene View or the [PREVIEW](#) area at the top of the window to select islands.
2. Toggle between **Add** mode (add to selection) and **Remove** mode (deselect).
 - You can quickly switch modes with a hotkey (default [R](#)).
3. Use the [Invert](#), [Select All](#), and [Clear](#) buttons for batch operations.
4. **Paint Mode:** Select "Paint" in the toolbar below the preview to directly paint masks using various tools.
 - **Tool Types:**
 - **Brush:** Freehand drawing.
 - **Rect:** Fills a rectangular area.
 - **Lasso:** Fills a hand-drawn enclosed area.
 - **Eraser:** Removes painted areas like a brush (island selection is not affected).
 - **Brush Size:** Adjusts thickness for Brush and Eraser tools (1–100px).

- **Undo / Redo / Clear:** Manage paint history or clear all hand-painted data.

STEP 3: Export

1. Confirm the resolution, invert mask option, and pixel margin in the **Quick Export** section.
2. Click **Save PNG** to export the mask image.

Section Descriptions

1. Target Model

- **Object:** The GameObject containing the target MeshRenderer or SkinnedMeshRenderer.
- **Target Material:** Focuses analysis on a specific submesh/material. Useful for models with overlapping UVs.
- **UV Channel:** Specifies the UV channel (UV0–UV7) used for analysis.
- **Create Work Copy / Remove Copy & Return:**
 - Creates a duplicate unaffected by mesh deformation (e.g., Modular Avatar).
 - Clicking "Remove Copy & Return" restores the session to the original focus.

1.5. Import Mask Image

Loads an existing mask image and treats its black pixels as hand-painted areas. This allows you to resume editing from a previously exported mask PNG instead of starting from scratch.

- **Image / Drop Area:** Drag and drop a **Texture2D** asset from the Project window onto the Object field, or select it with the picker.
- **Black Threshold (0–255):** Pixels whose luminance is below this value are treated as "painted" (black). Default is **128**.
 - **Lower values** (e.g., 32): Only very dark pixels are imported — useful when the mask has slight gray fringing.
 - **Higher values** (e.g., 200): Darker-gray pixels are also imported — useful when you want to capture semi-transparent or anti-aliased edges.
- **Load Mask button:** Applies the black regions of the image to the current hand-paint layer.
 - The import **merges** with existing paint data (OR semantics). Previously painted pixels are not erased.
 - The operation is **undoable** via the **Undo** button in the preview toolbar.

Tips: After loading, switch to Paint mode to fine-tune the imported areas with the brush or eraser.

2. Island Selection

- **Add / Remove:** Switches the basic behavior upon clicking.
- **Selection Count:** Displays the number of currently selected islands.
- **Batch Action Buttons:**
 - **Invert:** Selects unselected islands and deselects selected ones.
 - **Select All:** Selects all islands.
 - **Clear:** Deselects all islands.

3. PREVIEW

- Displays the analyzed UVs.
- **View Controls:** Use the mouse wheel to zoom and right-drag to pan.
- **size reset:** Resets the view to the initial position.
- You can also click directly here to select or deselect islands.

Hand-painting Details

In addition to island-based selection, you can draw freely on a pixel-by-pixel basis.

- **Switching Modes:** Select "Paint" in the toolbar below the preview to enter drawing mode (Switching back to "Select" will return to island selection).
- **Drawing Tools (Toolbar):**
 - **Brush:** Draws freehand lines.
 - **Rect:** Fills a rectangular region.
 - **Lasso:** Fills an enclosed area of any shape.
 - **Eraser:** Removes painted areas.
- **Undo / Redo:**
 - Each paint stroke can be individually reversed using **Undo / Redo**.
 - *Note: This is managed as a separate history from island selection changes.*
- **Merging with Island Selection:**
 - The final mask is the result of an **OR operation** between **[Selected Islands] + [Hand-painted areas]**.
 - This means that even if you use the eraser on an area covered by a selected island, it will still be included in the output unless you deselect the island itself.

Tips: It is most efficient to select the general shape using UV islands first, then use hand-painting for fine adjustments.

4. Quick Export

- **Resolution:** Output texture size.
- **Invert Mask:** Swaps black (opaque) and white (transparent) in the final output.
- **Pixel Margin:** Expands black areas by a few pixels to prevent bleeding near UV seams.

5. Output Settings (Advanced)

- **File Name:** Output filename (without extension).
- **Output Folder:** Destination folder. You can drag and drop folders or images into the frame.
- **Use Main Texture Folder:** Saves in the same location as the model's texture.
- **Also save inverted mask:** Outputs **_inv.png** alongside the normal mask.

Advanced Options

Scene Overlay

Display settings for the Scene View.

- **Draw on Top (X-Ray):** Always displays wires/seams in the foreground.
- **Thickness / Depth Offset:** Adjusts line weight and prevents flickering (z-fighting).

- **Color Settings:** Customizes colors for selected islands and seams.

Channel Write

Writes the mask only to specific RGB(A) channels.

- **Enabled:** Check to enable channel-specific writing.
- **Base PNG:** Specify an image to overwrite.

Vertex Color Bake

Bakes the selection mask as vertex colors into the mesh and saves it as a new asset.

Preferences

Global tool settings.

- **Language Mode:** Toggles between English and Japanese UI.
 - **Toggle Hotkey:** Key used for mode switching (default **R**).
 - **Auto Work Copy:** Toggles automatic creation of WorkCopy upon setting a target.
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Troubleshooting

Q. Clicking the mesh does not respond

- Ensure the target object (or WorkCopy) is correctly set.
- Ensure you are clicking the "front" face of the mesh (back faces are not detected).

Q. White bleeding occurs at mask boundaries

- Adjust the **Pixel Margin** value (recommended: 2px or more). This slightly expands the black area to fill UV gaps.